

Graphs of transformed hyperbolas

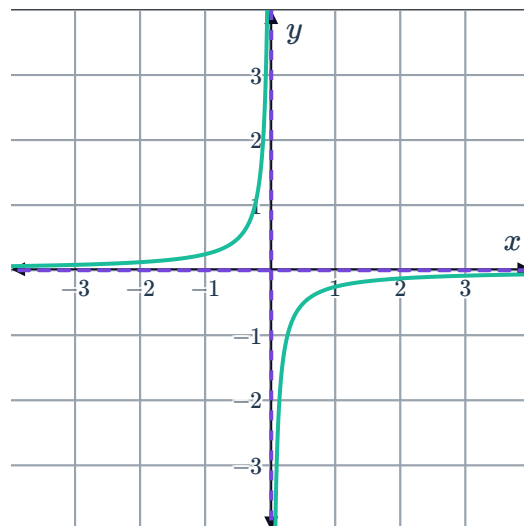


1

a

x	-3	-2	-1	1	2	3
y	$\frac{1}{12}$	$\frac{1}{8}$	$\frac{1}{4}$	$-\frac{1}{4}$	$-\frac{1}{8}$	$-\frac{1}{12}$

b



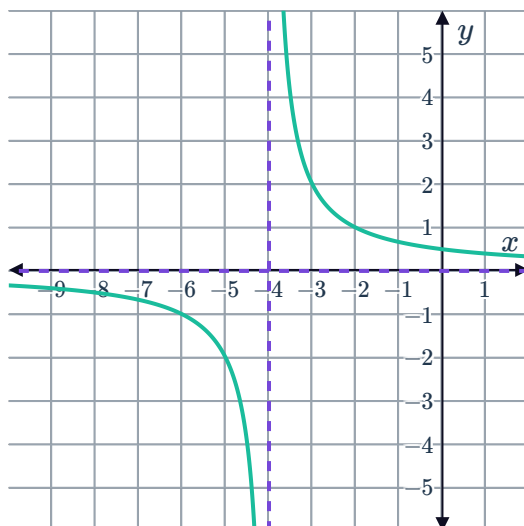
c 2 and 4

2

a $x = -4$

b $y = 0$

c



3

a

i No

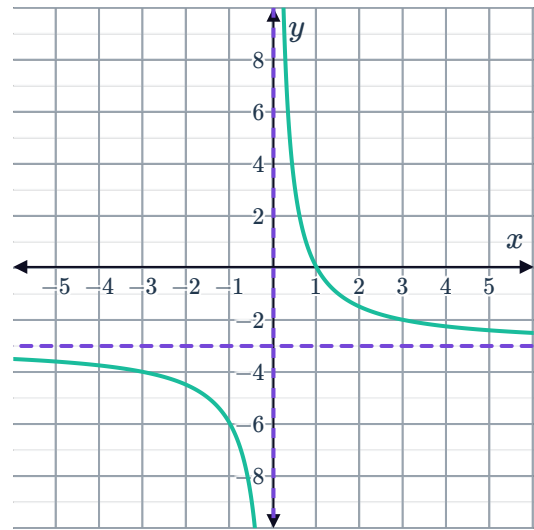
ii Yes

b $y = \frac{3}{x} - 3$

c $x = 0, y = -3$



d



4

a

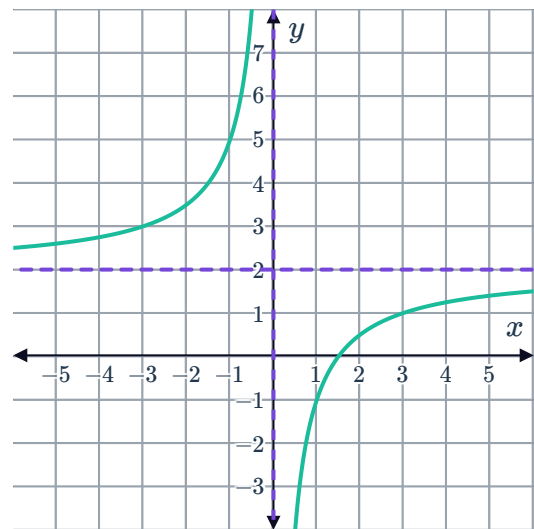
i Yes

ii No

c $x = 0, y = 2$

b $y = \frac{-3}{x} + 2$

d



5

a

i Yes

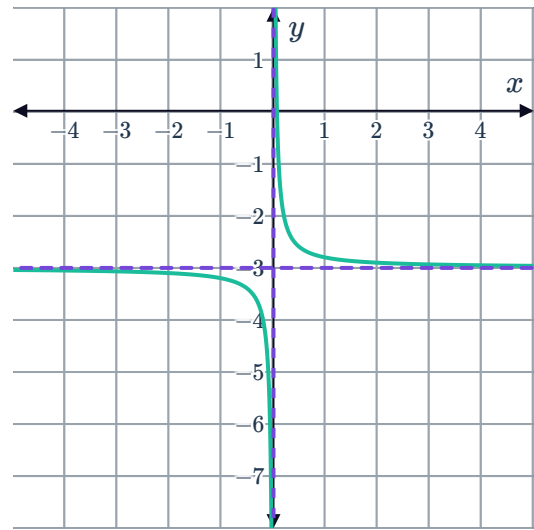
ii No

b $y = \frac{1}{x}$

c $x = 0, y = -3$



d



6

a

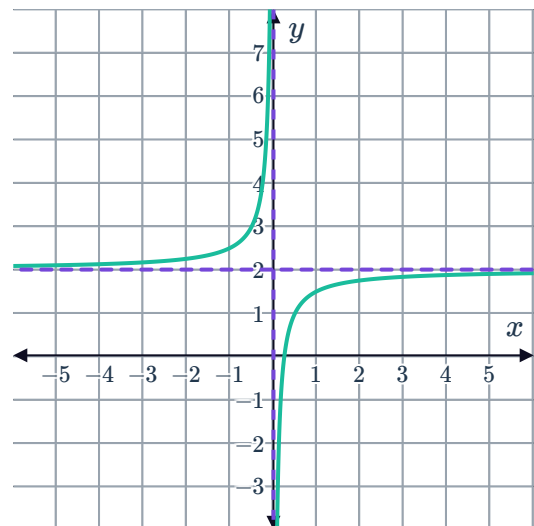
i Yes

ii No

c $x = 0, y = 2$

b $y = \frac{-1}{x}$

d

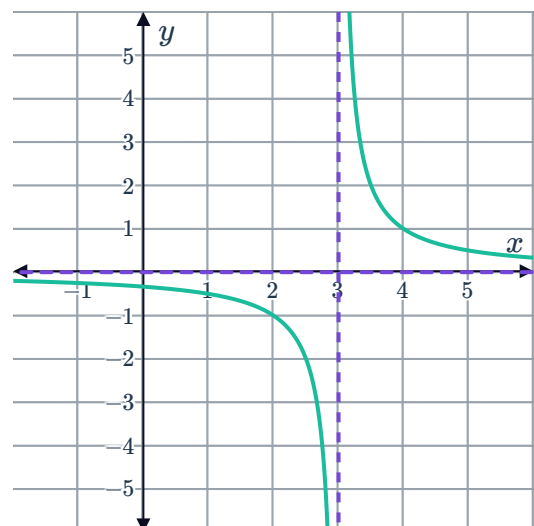


7

a

x	1	2	$\frac{5}{2}$	$\frac{7}{2}$	4	5
y	$-\frac{1}{2}$	-1	-2	2	1	$\frac{1}{2}$

b



8

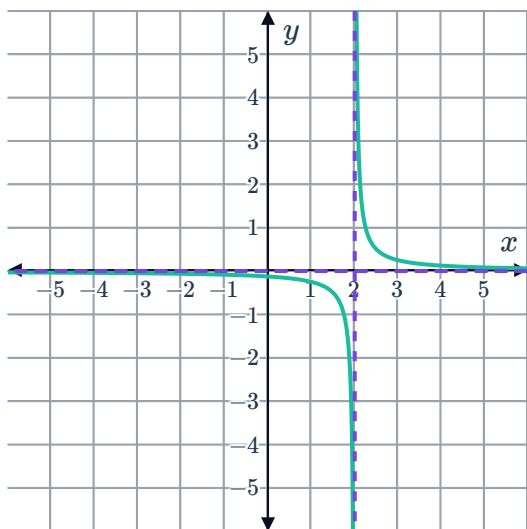


a

i B

ii $x = 2, y = 0$

iii

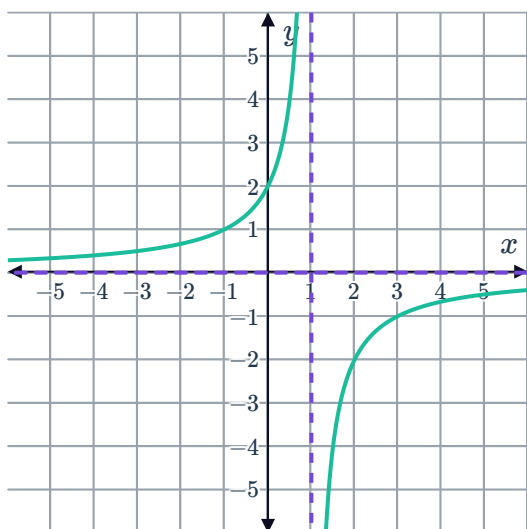


b

i A

ii $x = 1, y = 0$

iii



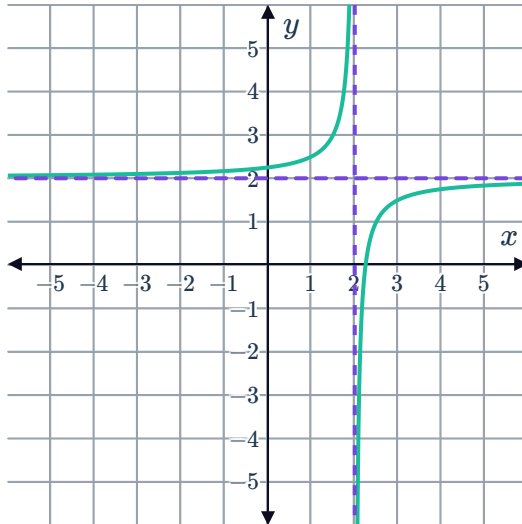
c

i A



ii $x = 2, y = 2$

iii



9

a 2

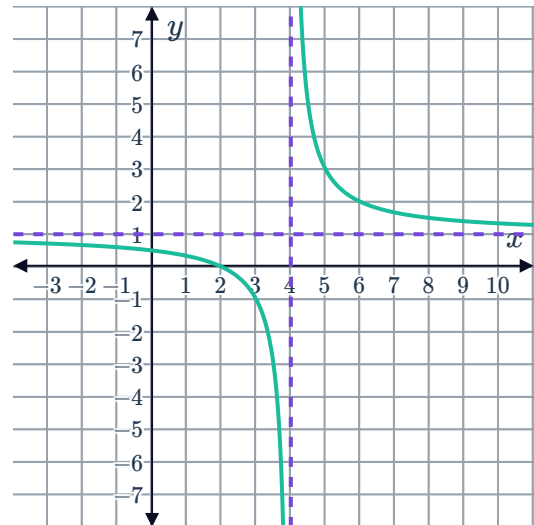
c $x = 4$

e $y = 1$

b $y = \frac{2}{x-4} + 1$

d 1

f



10

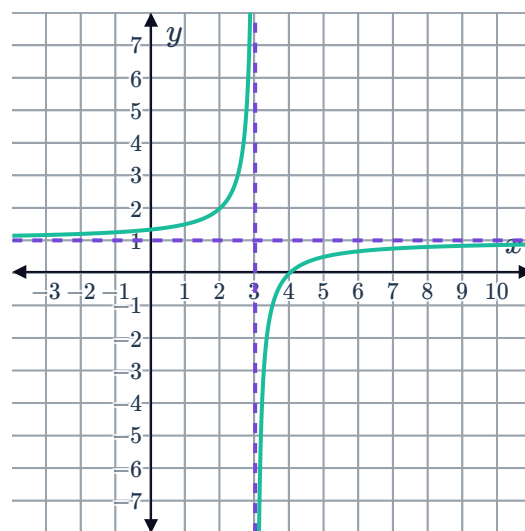
a 1

c $x = 3$

b $y = -\frac{1}{x-3} + 1$

d 1

e $y = 1$



11

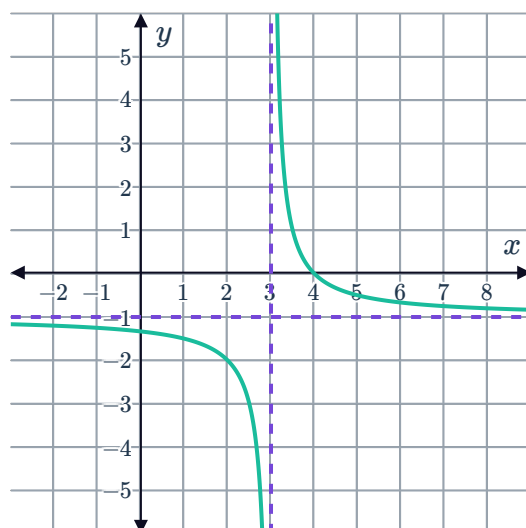
a $y = \frac{1}{x-3} - 1$

b $x = 3$

c -1

d $y = -1$

e



Transformations and domain and range

12

a $y = \frac{1}{x} + 4$

b $y = \frac{1}{x-7}$

13

a -3

b -6

c Translated 3 units down

14

a $x = 0$

c $x = 4$

e Translated 4 units right



b 2 and 4

d 1, 2 and 4

15

a Yes

b No

c Yes

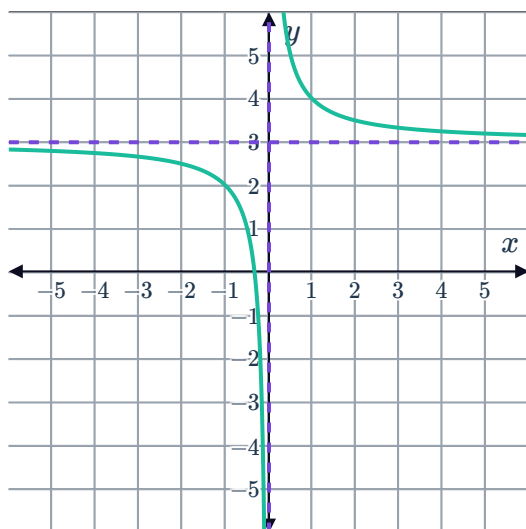
d No

16

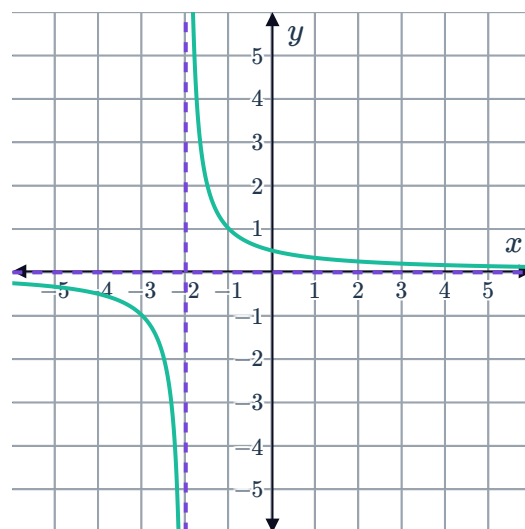
a Move the graph upwards by 3 units.

b Shift the graph to the left by 2 units.

c



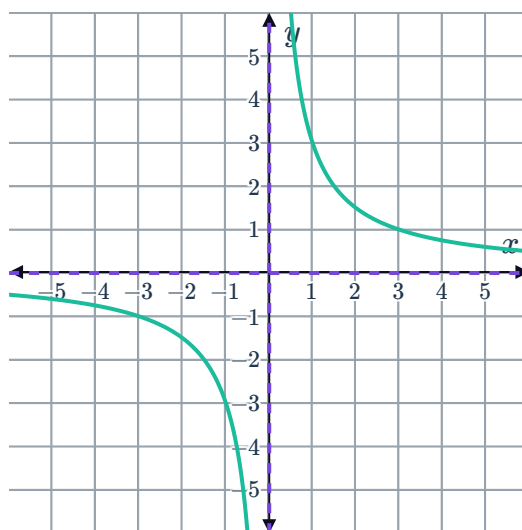
d



17

a Expansion away from the asymptotes by a factor of 3.

b



c $(-\infty, 0) \cup (0, \infty)$

d $(-\infty, 0) \cup (0, \infty)$

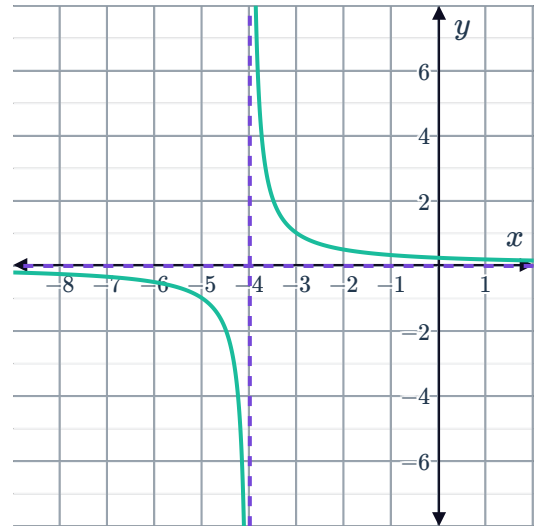
e Decreasing

18

- a Shift to the left by 4 units.



b



- c $(-\infty, -4) \cup (-4, \infty)$
e Decreasing

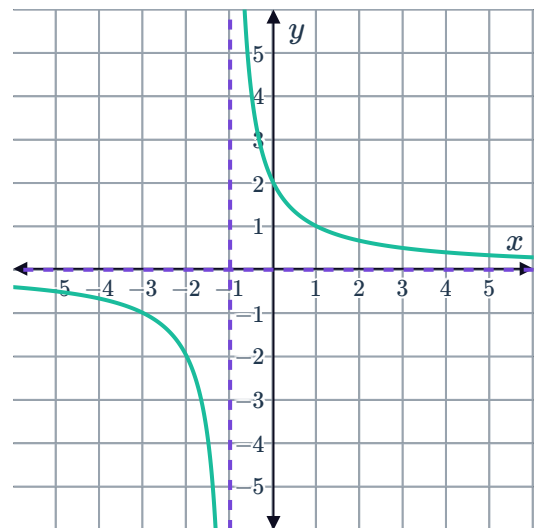
- d $(-\infty, 0) \cup (0, \infty)$

19

- a $(-\infty, -1) \cup (-1, \infty)$
c $x = \frac{2}{y} - 1$
e $y = 0$

- b $x = -1$
d $(-\infty, 0) \cup (0, \infty)$

f



20

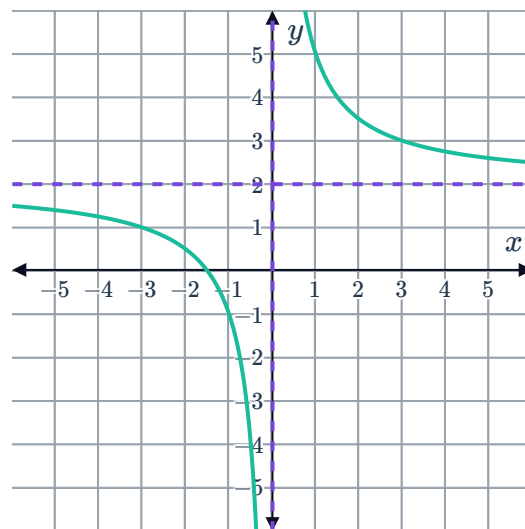
- a $x \in \mathbb{R}, x \neq 0$
c $x = \frac{3}{y-2}$

- b $x = 0$
d All real $y \neq 2$

e $y = 2$



f



21

a All real $x \neq 1$

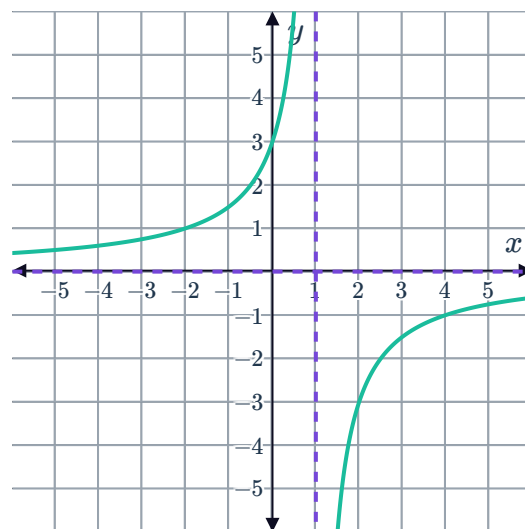
c $x = 1 - \frac{3}{y}$

e $y = 0$

b $x = 1$

d All real $y \neq 0$

f



22

a $a = -2$

c 0

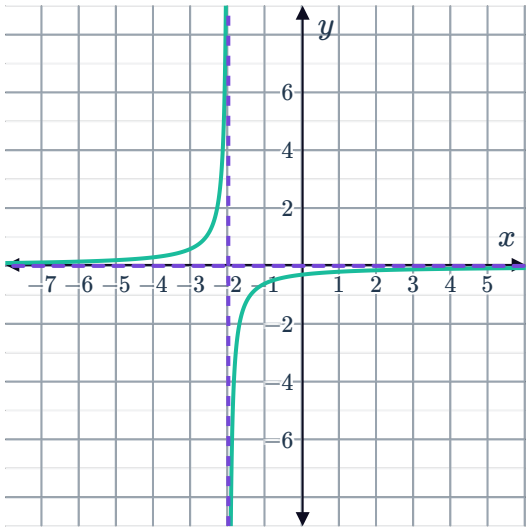
b $x = -2$

d $y = 0$

e 0



f



Applications

23

a $k = 1220$

b $t = 40$

24

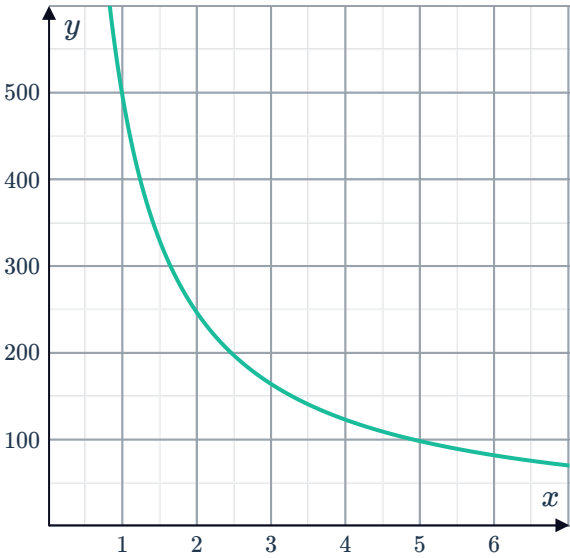
a \$245.00

b $y = \frac{490}{x}$

c

x	1	2	3	4	5	6
y	490	245	163.33	122.5	98	81.67

d



e Inverse Variation

$\frac{15}{2}$ units

